Restaurant order management

A SQL-based Restaurant Order Management System developed using MySQL to efficiently manage and analyze restaurant operations. This project demonstrates real-time database management concepts including menu management, customer orders, order details, billing, table reservations, customer records, and restaurant reporting.

Project Overview

The Restaurant Order Management System is a SQL-based database management project developed using MySQL to efficiently manage restaurant operations and perform business analytics. The system stores and manages data related to menu items, customers, customer orders, order details, billing, and table reservations. It demonstrates practical implementation of relational database concepts, SQL queries, and data analysis techniques for real-world restaurant management.

Project Objectives

- Design a normalized relational database for restaurant operations.
- Store and manage customer, menu, booking, billing, and order information.
- Perform data retrieval using SQL queries.
- Generate meaningful business reports through SQL analytics.
- Demonstrate the use of SQL joins, aggregate functions, subqueries, and ranking functions.
- Improve understanding of real-world database management systems.

Database Tables

1. Menu Items

Stores all food and beverage items available in the restaurant.

Features

- Item ID
- Item Name
- Category
- Price
- Availability Status

2. Customer Orders

Stores every order placed by customers.

Features

- Order ID
- Customer ID
- Table ID
- Order Date & Time
- Order Status

3. Order Details

Acts as the junction table between orders and menu items.

Features

- Order Detail ID
- Order ID
- Item ID
- Quantity
- Subtotal

4. Bills

Stores billing and payment information.

Features

- Bill ID
- Order ID
- Total Amount
- Payment Method
- Payment Status

5. Table Bookings

Stores reservation details.

Features

- Table ID
- Customer ID
- Booking Date & Time
- Number of Seats
- Booking Status

6. Customers

Stores customer information.

Features

- Customer ID
- Customer Name
- Phone Number
- Email Address
- Loyalty Points

Entity Relationship Highlights

The database consists of six interconnected tables using Primary Keys (PK) and Foreign Keys (FK).

Relationships

Customers → Customer Orders

- One customer can place multiple orders.

Customers → Table Bookings

- One customer can make multiple reservations.

Customer Orders → Order Details

- One order can contain multiple menu items.

Menu Items → Order Details

- One menu item can appear in many orders.

Customer Orders → Bills

- One completed order generates one bill.

Table Bookings → Customer Orders

- Orders may be associated with reserved tables.

SQL Concepts Used

This project demonstrates a wide range of SQL concepts including:

Basic SQL

- SELECT
- WHERE
- ORDER BY
- LIMIT

Filtering

- AND
- OR
- BETWEEN
- IN
- LIKE

Joins

- INNER JOIN
- Multiple Table JOINs

Aggregate Functions

- COUNT()
- SUM()
- AVG()
- MAX()
- MIN()

Grouping

- GROUP BY
- HAVING

Date & Time Functions

- NOW()
- DATE()
- TIMESTAMPDIFF()

Subqueries

Ranking Functions

- RANK()

Sorting

- ORDER BY ASC
- ORDER BY DESC

Learning Outcomes

Through this project, the following concepts were learned:

- Relational Database Design

- Database Normalization
- Primary and Foreign Key Relationships
- Writing Simple and Complex SQL Queries
- Data Analysis using SQL
- Aggregate Functions
- Multi-table Joins
- Subqueries
- Ranking Functions
- Business Reporting
- Real-world Database Management

Future Enhancements

The project can be enhanced by adding:

Employee Management Module

Kitchen Order Tracking

Inventory Management

Online Food Ordering Integration

Customer Feedback System

Discount & Coupon Management

Dashboard using Power BI/Tableau

Stored Procedures and Triggers

Role-based User Authentication

Real-time Analytics Dashboard

Conclusion

The Restaurant Order Management System successfully demonstrates the implementation of a relational database using MySQL to manage restaurant operations efficiently. By integrating menu management, customer information, table reservations, order processing, and billing, the project provides a structured solution for handling daily restaurant activities. The implementation of 40 SQL queries showcases essential SQL concepts ranging from basic retrieval to advanced analytics, making this project a practical demonstration of database design, data management, and business reporting skills applicable to real-world restaurant management systems.